

Introduction to TensorFlow and Keras

TensorFlow was developed by the Google Brain team and released in 2015 for performing large scale machine learning tasks. It is an open source and free software library, and is widely used to build, train and infer deep learning models. TensorFlow runs on a variety of platforms like Windows, Linux, macOS and even on Android. It is written in languages like C++, Python and CUDA. It allows training and deploying of ML and DL models very quickly, regardless of the language or platform.

Keras is a Python based open source library that runs on top of TensorFlow. Considered as a high-level API, it is easy to use and is also user friendly. It is more suitable for quick implementations and works well on small data sets. However, the number of projects available online in Keras is lesser as compared with TensorFlow.

Data preparation

The first step in building and deploying a CNN model is to prepare the data set. Accumulating a good quality data set has a major impact on the success of any deep learning model. There is a variety of data sets available for performing experiments using deep learning models. In this article, we have considered the Cat and Dog data set available on Kaggle to implement image classification with CNN using Keras. This data set contains 8007 training images and 2025 testing images. In this article, we have developed a classifier model that classifies a given picture as that of a dog or a cat.

Building a model

We have implemented the CNN using Google Colab – a browser based cloud platform which makes it possible to make complex models on large data sets using Jupyter Notebooks. The first step is to import the necessary packages from Keras. The following code can be used to import the required packages to implement CNN classification:

```
import os
from glob import glob
from tensorflow import keras
import numpy as np
from tensorflow.keras import layers
from keras.models import Sequential

from keras.layers import Activation, MaxPooling2D, Conv2D,
Lambda, Dropout, Dense, Flatten, Input
from keras.utils.vis_utils import plot_model
from keras.layers.normalization import BatchNormalization
```

We have created a simple CNN in Keras using a sequential API. Sequential API of Keras is basically a linear stack of multiple layers. We can add it layer by layer by using the `add()` method with parameters. Here, we have added three

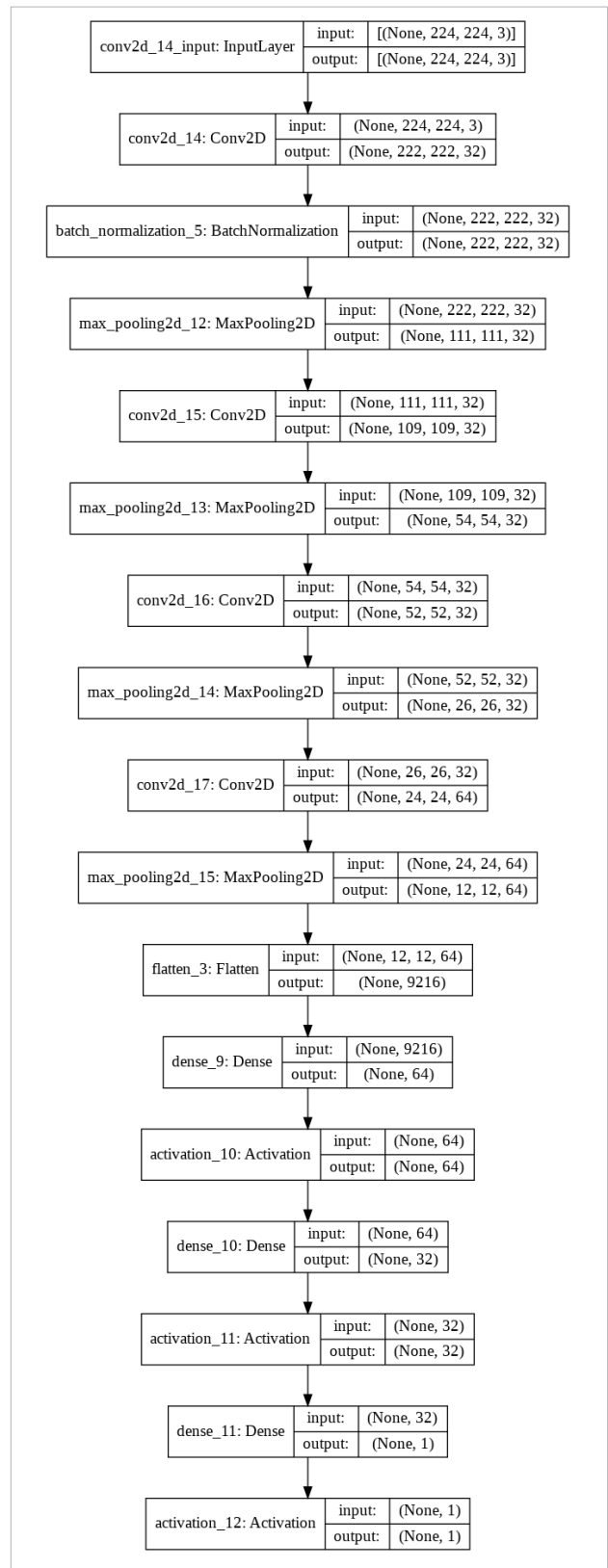


Figure 1: A plotted CNN model